

FORM TP 2011145



TEST CODE **02107032**

MAY/JUNE 2011

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 1 – PAPER 03/2

ALTERNATIVE TO INTERNAL ASSESSMENT EXAMINATION

2 hours

**Candidates are advised to use the first 15 minutes for
reading through this paper carefully.**

READ THE FOLLOWING DIRECTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. The use of silent non-programmable calculators is allowed.

1. (a) The labels from four bottles of solutions required for a laboratory experiment were accidentally removed. Each bottle contained one of the following: a protein solution, a sugar solution or distilled water. You are required to identify the contents of each bottle.

Materials

- I Eight test tubes in a test tube rack labelled A1, A2, B1, B2, C1, C2, D1, D2.
- II Four flasks labelled A, B, C and D respectively. The flasks contain samples of the solutions to be identified.

Procedure 1

- I Using the measuring cylinder labelled A, pour 2 cm³ of the solution from Flask A into each of the test tubes labelled A1 and A2.
- II Using the appropriately labelled measuring cylinder, repeat Step I for the solutions in Flasks B, C and D.
- III Using the measuring cylinder labelled KOH, add 2 cm³ of 5% KOH to the test tube labelled A1 and shake the test tube to mix.
- IV To the **same** test tube in III, add two drops of 1% copper sulphate solution and mix.
- V Observe for two minutes and note any colour change.
- VI Record your observation in Table 1 on this page.
- VII Repeat Steps III to VI using the test tubes labelled B1, C1 and D1.

TABLE 1: OBSERVATION

(i)

Solution	Colour Change
A1	
B1	
C1	
D1	

[3 marks]

- (ii) Name the food test being conducted in Procedure 1.

[1 mark]

GO ON TO THE NEXT PAGE

- (iii) State the conclusions that can be drawn from the observations in Table 1.

[2 marks]

(b) **Procedure 2**

- I Using the measuring cylinder labelled Benedict's solution, add 2 cm³ of Benedict's solution to the test tube labelled A2.
- II Shake the test tube and warm gently over the Bunsen burner.
- III Note the colour change and record your observation in Table 2 on this page.
- IV Repeat Steps I to III for each of the second test tubes labelled B2, C2 and D2.
- V Record your observations in Table 2 on this page.

TABLE 2: OBSERVATION

(i)

Solution	Colour Change
A2	
B2	
C2	
D2	

[3 marks]

- (ii) Name the nutrient being tested for, in Procedure 2.

[1 mark]

- (iii) State the conclusions that can be drawn from the observations in Table 2.

[1 mark]

- (iv) Which test tube MOST likely contains distilled water?

[1 mark]

GO ON TO THE NEXT PAGE

- (c) Some students in a class are asked to test fresh green beans for the presence of protein and reducing sugars.

- (i) Suggest what should be done to the beans **before** the food tests are carried out.

[3 marks]

- (ii) State ONE possible source of error in this investigation.

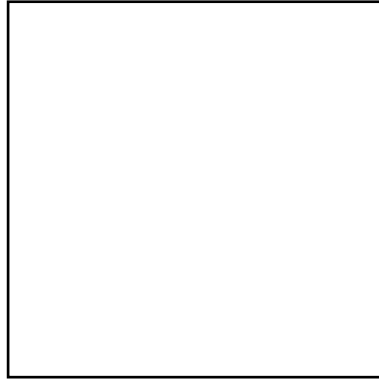
[1 mark]

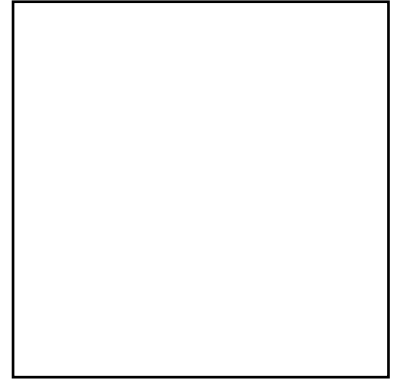
Total 16 marks

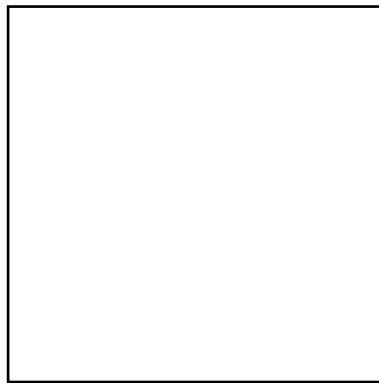
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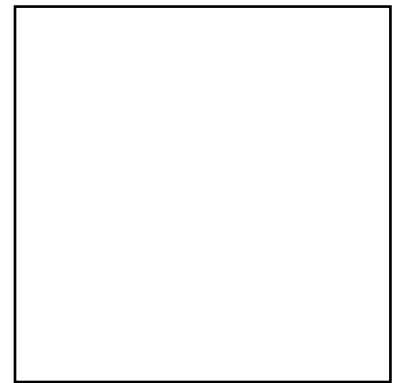
2. (a) Specimen A is a stained longitudinal section through a root tip showing cells in various stages of mitosis.

- (i) Draw a different stage of mitosis in EACH of the four boxes below. Write the name of the stage under EACH box. **DO NOT label the drawings.**









[6 marks]

- (ii) For EACH of the stages drawn, write the number 1, 2, 3 or 4, next to the name of the stage, to indicate the correct sequence in which they occur in the cycle. **[2 marks]**
- (iii) Name ONE structure that is seen in some of the dividing cells in Specimen A but will NOT be seen in dividing **animal** cells viewed under the same magnification.

_____ **[1 mark]**

GO ON TO THE NEXT PAGE

- (b) In the sweet pea plant, flower colour is determined by two allelomorphic pairs of genes (R, r and S, s). Purple flowers are produced if at least one dominant allele for each gene is present in the flower. All other genotypes are white. A plant that is homozygous for purple colour is crossed with a plant having the genotype, RrSs.

- (i) State the genotype of the flower that is homozygous for purple colour.

[1 mark]

- (ii) Complete the Punnett square below to show the expected genotypes of the offspring of the cross.

[4 marks]

- (iii) State the phenotype of the offspring produced by the cross.

[1 mark]

- (iv) Suggest how the results would have been different if both parents were heterozygous for purple color.

[1 mark]

Total 16 marks

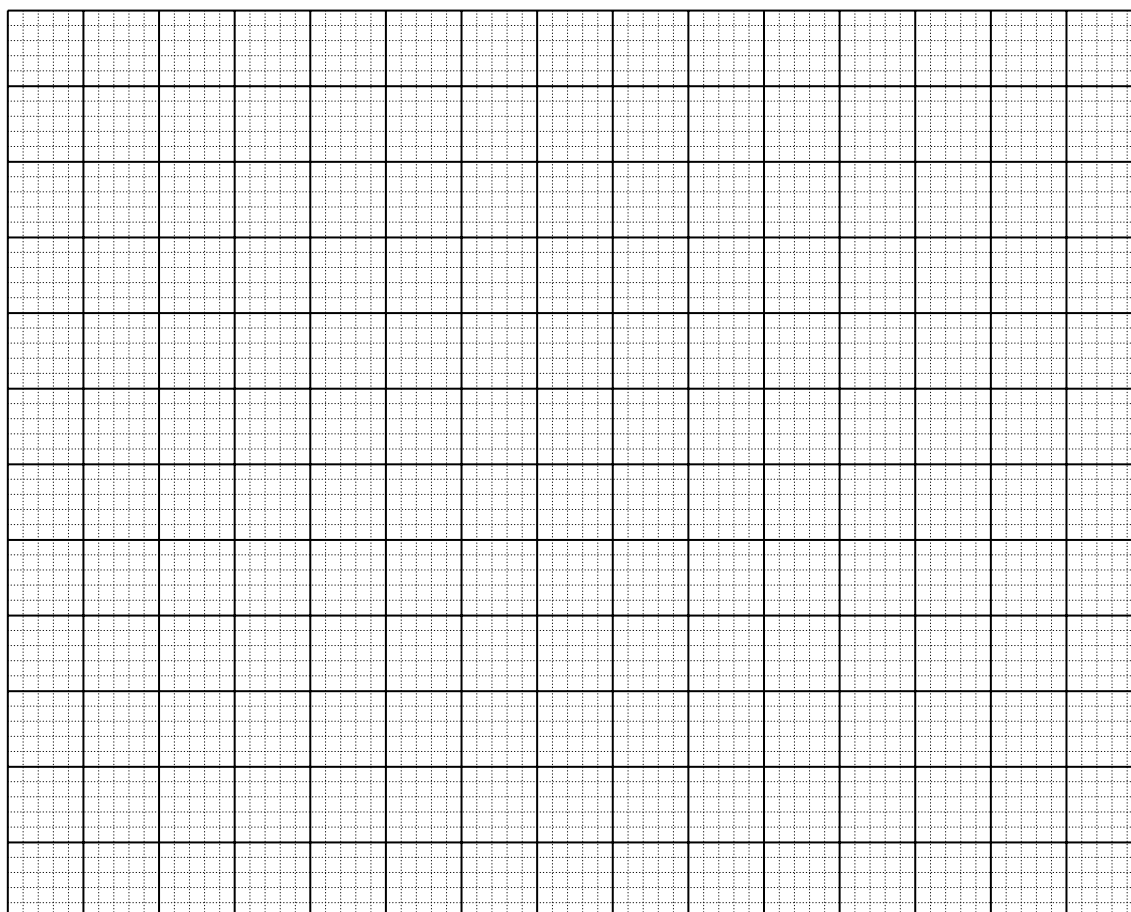
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3. (a) Table 3 presents data on the percentage of women in a sample who became pregnant while using specific contraceptive methods. The data was collected after the first 6 months of use and again after 12 months of use.

**TABLE 3: PERCENTAGE OF WOMEN IN SAMPLE WHO BECAME PREGNANT,
BY CONTRACEPTIVE METHOD AND DURATION OF USE**

Contraceptive Method	Duration of Use	
	6 months	12 months
Pill	3.3	7.0
Diaphragm	5.8	9.5
Male condom	5.1	9.2
Periodic abstinence	13.8	20.5

- (i) On the grid provided below, draw a bar chart to illustrate the data in Table 3.



[6 marks]

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- (ii) Identify THREE trends shown in the data.

[3 marks]

- (b) Figure 1 illustrates the changes in total fertility rates for three countries in the Caribbean between 1970 and 1995.

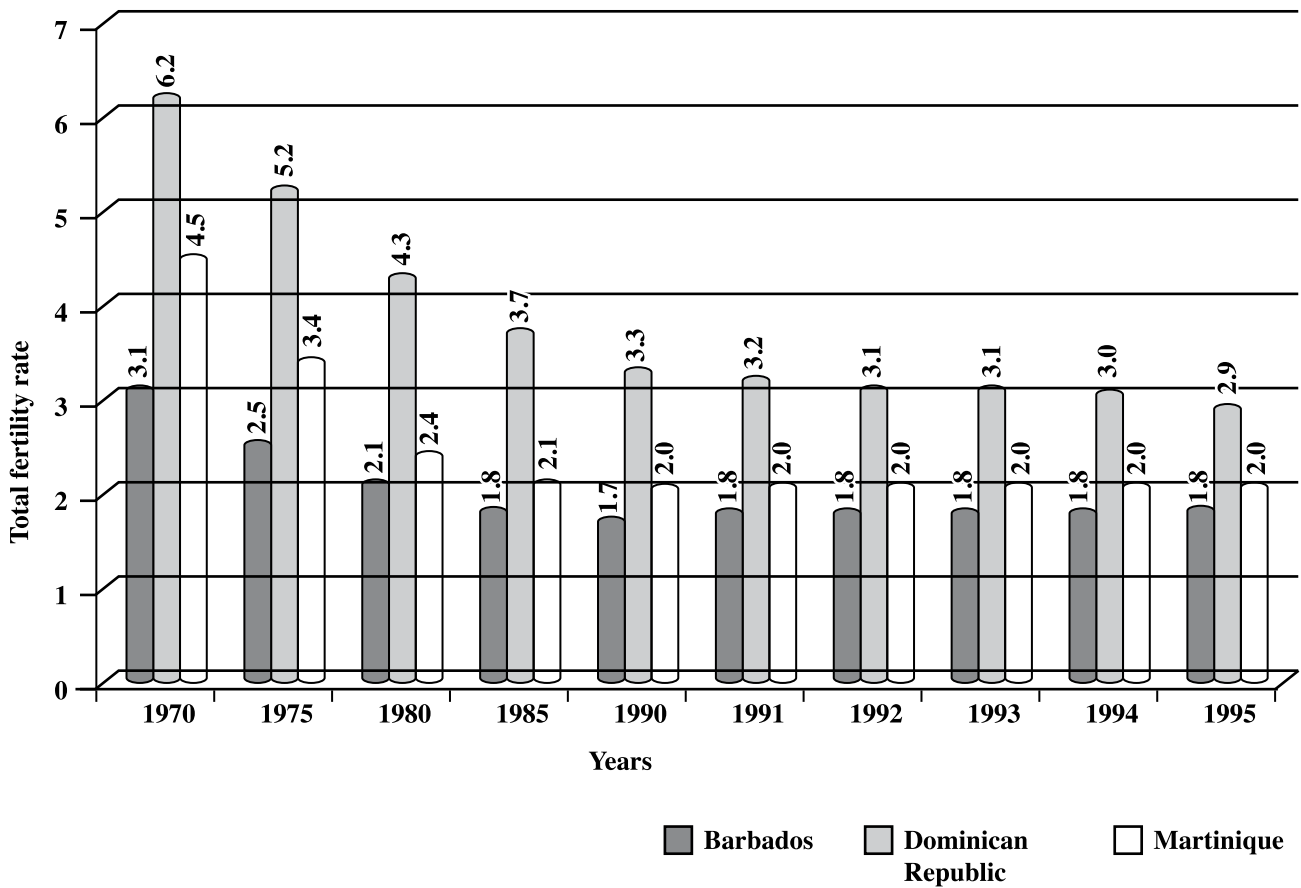


Figure 1. Changes in total fertility rates in Barbados, the Dominican Republic and Martinique

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- (i) Complete Table 4 using the data in Figure 1.

**TABLE 4: TOTAL FERTILITY RATES FOR BARBADOS,
THE DOMINICAN REPUBLIC AND MARTINIQUE**

Country	Total Fertility Rate	
	1970	1995
Barbados		
Dominican Republic		
Martinique		

[3 marks]

- (ii) Identify TWO general trends shown by the data over the period 1970 – 1995.

[2 marks]

- (iii) Suggest the total fertility rate for the Dominican Republic for the year 1996.

[1 mark]

- (iv) Comment on the implications for the Caribbean of the trends in total fertility rates shown in Figure 1.

[1 mark]

Total 16 marks